Discrete random variables

POLI 210 - Seminar 4

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Today's questions

- How does an experiment produce a numerical outcome?
- How does probability on outcomes become a distribution over numbers?
- What do a distribution's mean and variance tell us?
- How can we compare exact calculations with simulated experiments?

Reading: Speegle and Clair, *Probability, Statistics, and Data*, Chapter 3.

Outcomes, events, and random variables

For two coin flips, the sample space is

$$S = \{TT, HT, TH, HH\}.$$

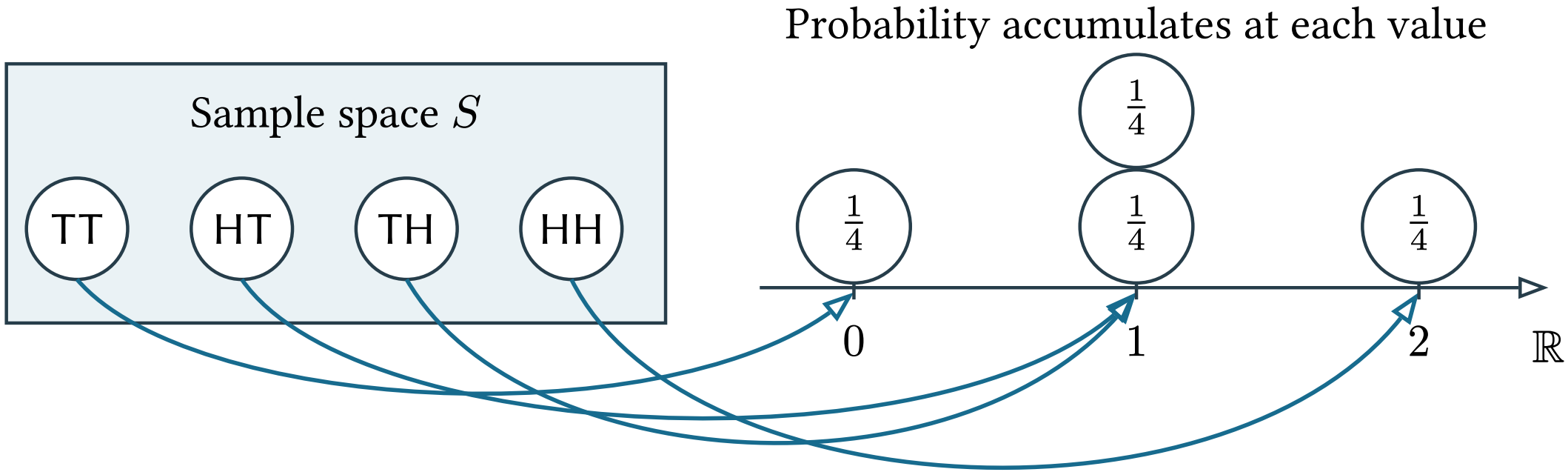
An **outcome** is one complete result, such as HT.

An **event** is a set of outcomes, such as $\{HT, TH\}$.

A **random variable** is a numerical function of the outcome:

$$X : S \to \mathbb{R}, \qquad X(s) = \text{number of heads in } \ s.$$

Mapping outcomes to numbers



$X(s)$ counts heads

$p(1) = P(X = 1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

Independent fair flips give each outcome probability $1/4$.

Recreated after Matthew Blackwell's slide 11, used with permission.

The induced probability distribution

Both HT and TH map to 1, so

$$\{X = 1\} = \{HT, TH\}, \qquad P(X = 1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

The **probability mass function** gives $p(x) = P(X = x)$.

x	0	1	2
$p(x)$	$1/4$	$1/2$	$1/4$

Equally likely outcomes can produce values with unequal probabilities.

A different summary of the same experiment

Define $W = 1$ when the two flips match and $W = 0$ otherwise.

Outcomes	W	Total probability
TT, HH	1	$1/2$
HT, TH	0	$1/2$

The experiment and its outcome probabilities stay the same.

We change the numerical function that summarizes the outcome.

What makes a random variable discrete?

A discrete random variable has **finitely or countably many** possible values.

- A count of treated people takes values $0, 1, 2, \ldots$.
- The fraction of heads in two flips takes values $0, 1/2, 1$.
- A waiting time measured in whole calls may have no finite upper bound.

The values need not be integers.

Its PMF satisfies $p(x) \geq 0$ and $\sum_x p(x) = 1$.

A voter-mobilization experiment

Three people are eligible for a mobilization message.

Independently assign each person to treatment T or control C with probability $1/2$.

Let X count the people assigned to treatment.

Question: What values can X take? Are they equally likely?

This hypothetical assignment mechanism describes who receives the message. It makes no claim about the message's effect.

The treatment-count PMF

Outcomes	x	Count: $\binom{3}{x}$	Prob. per outcome	$p(x)$
CCC	0	1	$1/8$	$1/8$
TCC, CTC, CCT	1	3	$1/8$	$3/8$
TTC, TCT, CTT	2	3	$1/8$	$3/8$
TTT	3	1	$1/8$	$1/8$

Counting connection: Choose which x of the three people receive treatment.

In R, `choose(3, 0:3)` gives `1, 3, 3, 1`.

$$P(\text{both groups exist}) = P(X = 1) + P(X = 2) = \frac{3}{4}.$$

Catch: Independent coin flips do not guarantee fixed group sizes.

Enumeration in R

```
assignments <- expand.grid(person1 = c("C", "T"),
                           person2 = c("C", "T"),
                           person3 = c("C", "T"))
number_treated <- rowSums(assignments == "T")
table(number_treated) / nrow(assignments)
```

```
number_treated
    0     1     2     3
0.125 0.375 0.375 0.125
```

Each row is one complete assignment. We count outcomes exactly here.

Live demo 1: Inspect the rows and their numerical summaries.

Simulating the assignment mechanism

```r
set.seed(92301)
N <- 20000L
X <- replicate(N, {
  assignment <- sample(c("C", "T"), 3, replace = TRUE)
  sum(assignment == "T")
})
head(X)
```

```
[1] 0 0 0 2 2 1
```

One repetition represents one experiment with **three people**.

Each entry of `X` records one treatment count.

Live demo 2: Predict which counts will occur most often.

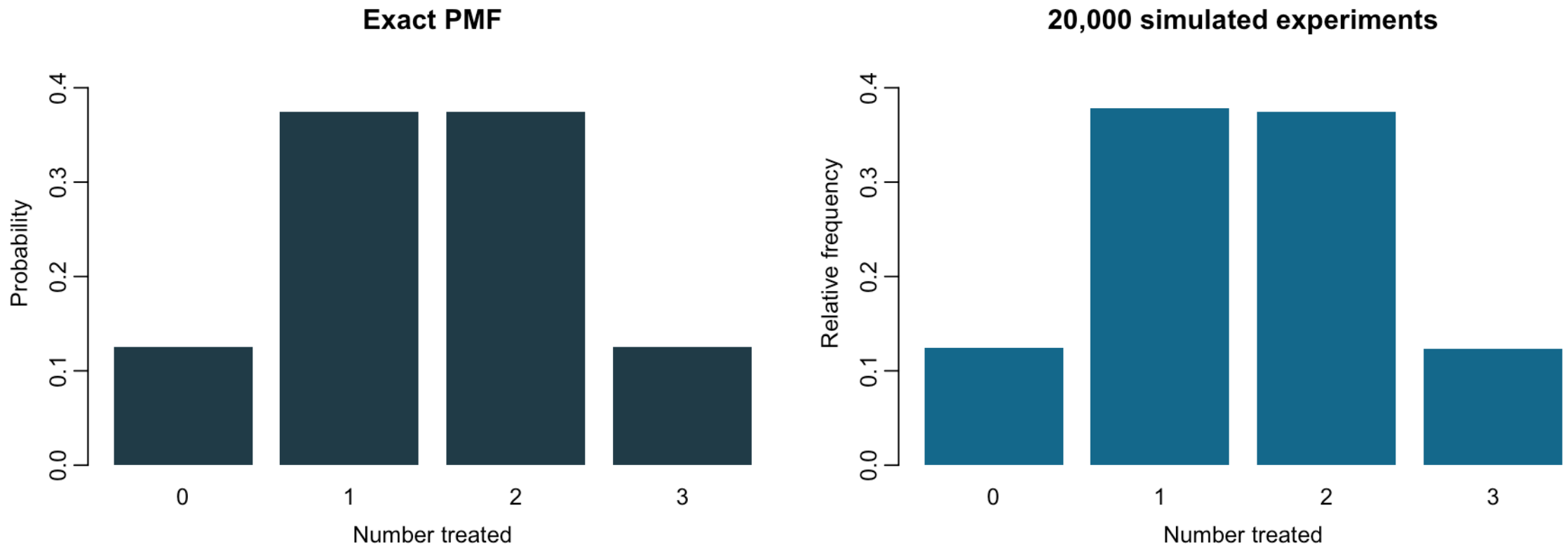
Simulating directly from the PMF

```
set.seed(92302)
X_direct <- sample(0:3, 20000, replace = TRUE,
                   prob = c(1, 3, 3, 1) / 8)
X_binom <- rbinom(20000, size = 3, prob = .5)
```

These commands produce counts directly, without retaining individual assignments.

Catch: `sample(0:3, 20000, replace=TRUE)` would give the four counts equal probabilities. That would describe a different model.

Exact and simulated distributions



More repetitions improve the approximation of the fixed PMF.

They do not reduce variation across individual three-person experiments.

Expected value

For a discrete random variable with finite expectation,

$$E[X] = \sum_x x\, p(x).$$

The expectation is a probability-weighted average of the possible values.

Think of probability weights placed on a number line. The expectation is their balance point.

Question: Must that balance point be a value X can actually take?

The expected treatment count

x	$p(x)$	$xp(x)$
0	$1/8$	0
1	$3/8$	$3/8$
2	$3/8$	$6/8$
3	$1/8$	$3/8$

$$E[X] = 0\left(\frac{1}{8}\right) + 1\left(\frac{3}{8}\right) + 2\left(\frac{3}{8}\right) + 3\left(\frac{1}{8}\right) = 1.5.$$

Every actual experiment treats an integer number of people.

An average and an event probability

```
mu <- sum(xs * pmf)
c(exact_mean = mu, simulated_mean = mean(X),
  probability_all_same = mean(X == 0 | X == 3))
```

```
          exact_mean       simulated_mean probability_all_same
             1.50000              1.49750              0.24775
```

`mean(X)` averages treatment counts.

`mean(X == 0 | X == 3)` averages an event indicator.

Live demo 3: Compare the two quantities and explain their units.

Why repeated simulation works

With independent repetitions of the same model and finite expected absolute value, the **Law of Large Numbers** links sample averages to expectations.

For an indicator D with $P(D = 1) = p$,

$$E[D] = 0(1 - p) + 1p = p.$$

This is why the average of a logical comparison estimates a probability.

Catch: A larger simulation need not be closer on every run.

Break

Before we pause:

Why can equally likely experiment outcomes produce a nonuniform PMF?

After the break: polling, transformations, and variation.

Bernoulli trials and binomial counts

A Bernoulli variable is 1 for an event labeled success and 0 otherwise.

Suppose n trials are independent and each has success probability p. Let X count successes. Then

$$X \sim \text{Binom}(n, p),$$

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, \ldots, n.$$

The assumptions are part of the model. A count alone need not be binomial.

Where the binomial formula comes from

For exactly x successes in n trials:

1. Choose the x successful positions: $\binom{n}{x}$ choices.
2. Each sequence has probability $p^x(1-p)^{n-x}$.
3. Add the probabilities of those disjoint sequences.

Our experiment has $X \sim \text{Binom}(3, 1/2)$:

$$p(x) = \binom{3}{x}(1/2)^3.$$

A polling model

Suppose a fixed **46%** of a large voter population supports Peyton. A poll samples **300 voters**. Let K count Peyton supporters.

$$K \sim \text{Binom}(300, .46), \qquad E[K] = 138.$$

Sampling without replacement is approximately binomial when the poll is a small share of the population.

What is the probability Peyton has a **strict majority in the sample**?

Hypothetical setup from PSD Example 3.21. We do not model turnout or nonresponse.

The probability of a sample lead

A strict majority requires $K > 150$. A 150-150 tie does not count.

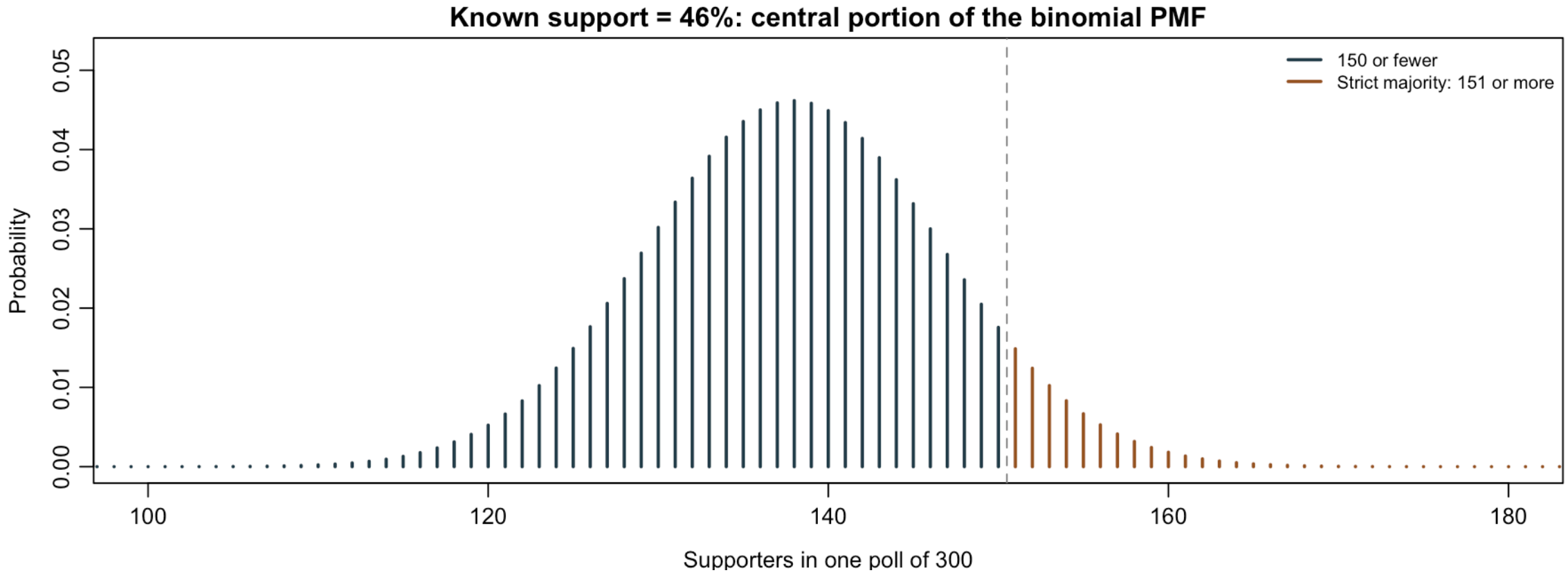
```
pbinom(150, size = 300, prob = .46, lower.tail = FALSE)
```

```
[1] 0.07398045
```

$$P(K > 150) \approx .074.$$

This probability describes repeated polls under assumed population support. It is not a posterior probability of winning the election.

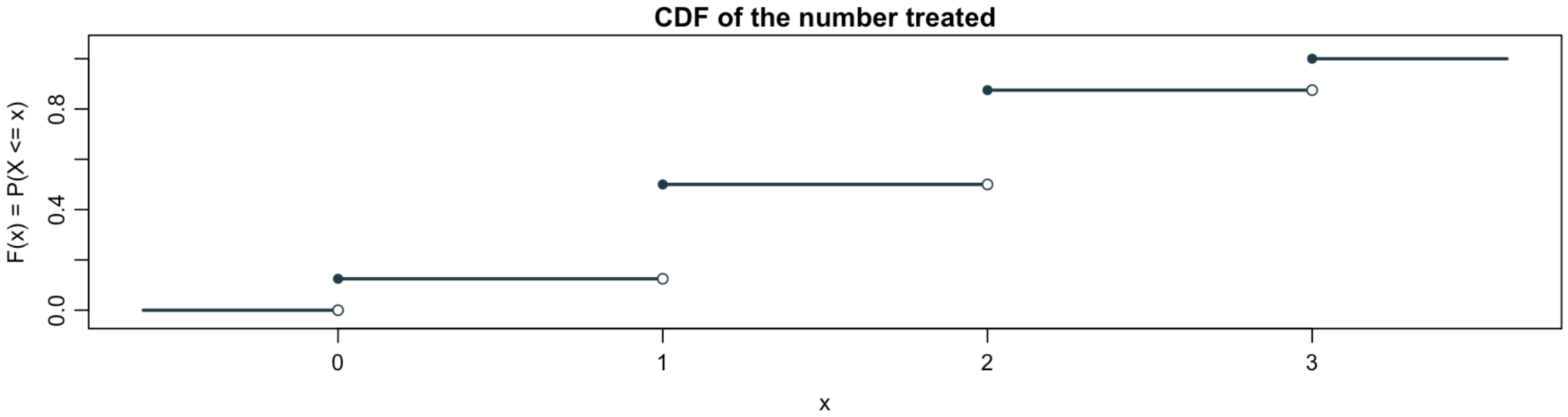
The tail of the polling distribution



The orange masses represent polls with 151 or more Peyton supporters.

Cumulative distribution functions

$$F(x) = P(X \leq x) = \sum_{u \leq x} p(u).$$



For the treatment count, $F(1) = F(1.4) = 1/2$.

Live demo 5: Inspect the filled and open endpoints.

Boundary conditions and R prefixes

Command	Question it answers
`dbinom(x, n, p)`	$P(K = x)$
`pbinom(x, n, p)`	$P(K \leq x)$
`rbinom(N, n, p)`	Generate N counts

`lower.tail=FALSE` gives $P(K > x)$.

For integer-valued K,

$$P(K \geq k) = P(K > k - 1).$$

Catch: $P(X < 1) = 1/8$, while $P(X \leq 1) = 1/2$ in our treatment example.

Simulating many polls

```
set.seed(92304)
polls <- rbinom(N, size = 300, prob = 0.46)
c(simulated_mean = mean(polls),
  simulated_probability_lead = mean(polls > 150),
  exact_probability_lead = poll_lead)
```

```
            simulated_mean simulated_probability_lead
              137.97330000                 0.07165000
    exact_probability_lead
                0.07398045
```

20,000 counts repeated polls. **300** counts voters within each poll.

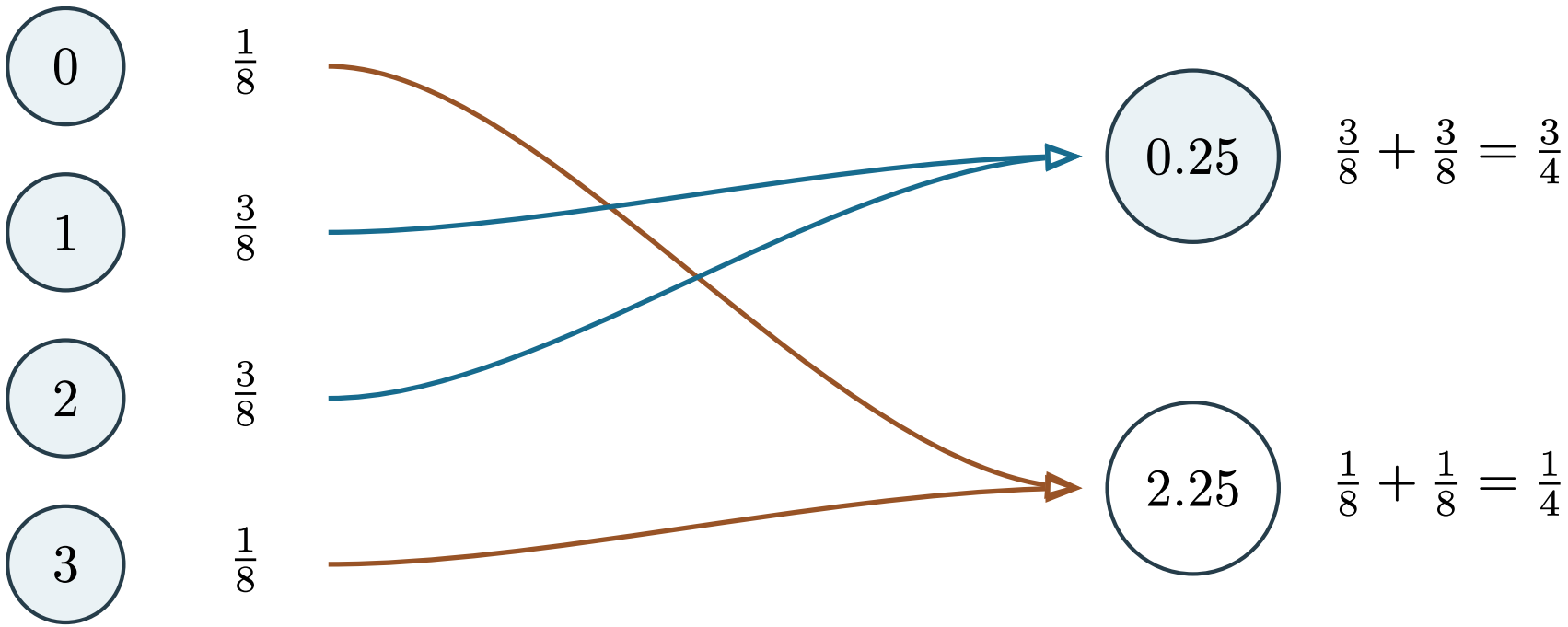
Live demo 4: Compare a strict lead with a tie or lead.

Transforming a random variable

X: number treated

$$Y = (X - 1.5)^2$$

Map values; add their probabilities



$$E[Y] = 0.25\left(\tfrac{3}{4}\right) + 2.25\left(\tfrac{1}{4}\right) = 0.75$$

Expectation after a transformation

Let $Y = g(X) = (X - 1.5)^2$. We can average using the original probabilities:

$$E[g(X)] = \sum_x g(x)p(x) = .75.$$

```
Y <- (X - 1.5)^2
c(prob_Y_0.25 = mean(Y == .25), prob_Y_2.25 = mean(Y == 2.25),
  mean_Y = mean(Y))
```

```
prob_Y_0.25 prob_Y_2.25      mean_Y
    0.75225     0.24775     0.74550
```

Catch: $E[(X - 1.5)^2] = .75$, but $(E[X] - 1.5)^2 = 0$.

Live demo 6: Apply the function to every simulated count.

Variance and standard deviation

For $\mu = E[X]$,

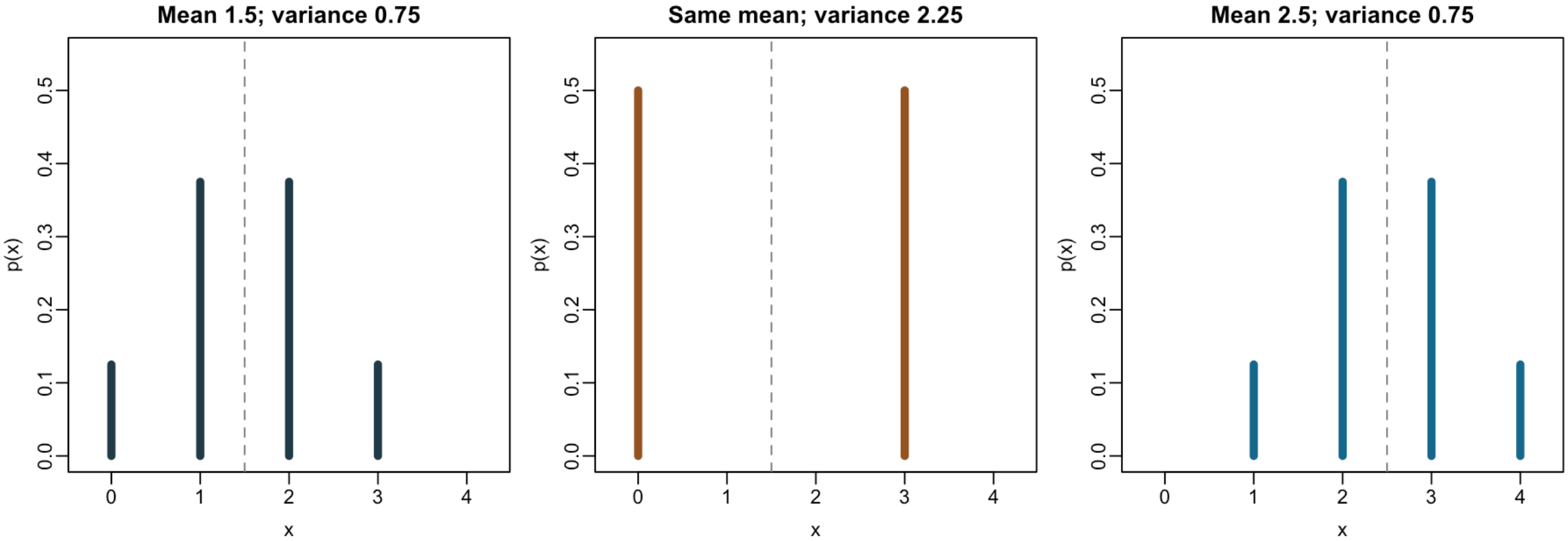
$$\mathrm{Var}(X) = E[(X-\mu)^2] = \sum_x (x-\mu)^2 p(x).$$

$$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

Signed deviations average to zero. Squared deviations measure spread.

Our treatment count has variance $.75$ and standard deviation $\sqrt{.75} \approx .866$ people.

Center and spread



Middle: one shared coin assigns everyone to the same group.

Right: adding 1 shifts the mean and preserves the variance.

Variance in R

```
mu <- sum((0:3) * c(1, 3, 3, 1) / 8)
c(model_variance = sum(((0:3) - mu)^2 * c(1, 3, 3, 1) / 8),
  mean_squared_deviation = mean((X - mu)^2),
  sample_variance = var(X))
```

```
        model_variance mean_squared_deviation        sample_variance
              0.750000               0.745500               0.745531
```

`var(X)` uses the sample mean and denominator $N - 1$.

`mean((X-mu)^2)` uses the known model mean. Their finite-simulation values can differ slightly.

Linearity of expectation

For finite expectations,

$$E[aX + bY + c] = aE[X] + bE[Y] + c.$$

Independence is unnecessary for this rule.

If D_i indicates success on trial i, with $P(D_i = 1) = p$,

$$E\left[\sum_{i=1}^{n} D_i\right] = \sum_{i=1}^{n} E[D_i] = np.$$

This gives the binomial mean.

Variance under shifting and scaling

$$\mathrm{Var}(aX + b) = a^2 \ \mathrm{Var}(X).$$

Adding b shifts the center without changing distances from it.

Multiplying by a multiplies each deviation by a, and its square by a^2.

An equivalent formula is

$$\mathrm{Var}(X) = E[X^2] - E[X]^2.$$

Independence and sums

For independent variables with finite variances,

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y).$$

A Bernoulli variable has variance $p(1 - p)$, so a binomial count has

$$\operatorname{Var}(X) = np(1 - p).$$

Catch: Reusing the same variable creates dependence:

$$\operatorname{Var}(X + X) = 4 \operatorname{Var}(X).$$

A new count versus a copy

```
set.seed(92305)
Z <- rbinom(N, size = 3, prob = .5) # independent of the draws in X
c(var_sum_independent = var(X + Z),
  sum_of_variances = var(X) + var(Z),
  var_sum_copied = var(X + X),
  four_times_variance = 4 * var(X))
```

```
var_sum_independent    sum_of_variances      var_sum_copied four_times_variance
           1.497760            1.492829            2.982124            2.982124
```

A fresh draw and a copied value give different joint behavior.

Live demo 7: Predict both answers before running the code.

More voters versus more simulations

If $K \sim \text{Binom}(n, p)$ and $\hat{p} = K/n$,

$$E[\hat{p}] = p, \qquad \text{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

Increasing **voters per poll** reduces variation in the sample proportion.

Increasing **simulated polls** improves our numerical description of that distribution.

These are different changes to the simulation.

Waiting for a volunteer

Suppose independent recruitment calls each have probability $.2$ of success. Let G count **unsuccessful calls before the first volunteer agrees**.

$$P(G = g) = (.8)^g(.2), \qquad g = 0, 1, 2, ...$$

$$G \sim \text{Geom}(.2), \qquad E[G] = \frac{1 - .2}{.2} = 4.$$

The total number of calls is $C = G + 1$, with $E[C] = 5$.

PSD and R count failures. Some other books count total attempts.

More than five calls

The first five calls must fail:

$$P(C > 5) = P(G \geq 5) = .8^5 = .32768.$$

```
pgeom(4, prob = .2, lower.tail = FALSE)
```

```
[1] 0.32768
```

Why 4 rather than 5? `lower.tail = FALSE` gives $P(G > q)$. Since G counts failures, we need $P(G \geq 5) = P(G > 4)$.

Using `pgeom(5, ...)` would give $P(G \geq 6)$, excluding exactly five failures.

Catch: The geometric success here is a volunteer agreeing. Always state which event stops the waiting time.

A geometric simulation

```
set.seed(92307)
G_direct <- rgeom(20000, prob = .2)
c(mean_failures = mean(G_direct), p_over_five = mean(G_direct >= 5))
```

```
mean_failures   p_over_five
       4.0086         0.3258
```

Live demo 8: Use a `while` loop to generate the calls themselves.

Different contact probabilities or a finite contact list may invalidate this simple model.

Other discrete models

Model	Example mechanism	R sampler
Hypergeometric	Count red marbles in a fixed draw without replacement	`rhyper()`
Poisson	Count arrivals in a fixed interval under a constant-rate model	`rpois()`
Negative binomial	Count failures before the rth success	`rnbinom()`

The mechanism determines whether the model is appropriate.

These are a brief reading guide to PSD §3.6. The companion has optional hypergeometric and Poisson examples.

Questions to take away

- What distinguishes an outcome, a random variable, and its distribution?
- How do multiple outcomes contribute to one PMF value?
- Why can a mean be impossible in any single experiment?
- Which expectation and variance rules require independence?
- What changes when we increase simulation repetitions?

Next week: continuous random variables, densities, and the normal model.

Sources

Speegle and Clair, *Probability, Statistics, and Data*, Chapter 3. The polling setup follows Example 3.21.

Matthew Blackwell, *Gov 2000: Random Variables and Probability Distributions* (2016), used with permission. The mapping graphic and assignment example adapt slides 11 and 13-15. The diagrams are editable Typst/CeTZ figures.

R distribution documentation. All computations use base R.